

Commerce XI Maths App

Miscellaneous Exercise 3

Detailed Answers with Explanations and Steps

Mathematics & Statistics, Part 1, Unit 3 - Complex Numbers | Total questions: 26

Question 1 - Q1

Find $(i^{592} + i^{590} + i^{588} + i^{586} + i^{584}) / (i^{582} + i^{580} + i^{578} + i^{576} + i^{574})$

Options

- A. -1
- B. 1
- C. i
- D. $-i$

Correct answer: -1

Explanation and steps

Reduce each exponent modulo 4 using $i^4 = 1$. Simplify numerator and denominator separately; their quotient is -1.

Question 2 - Q2

Find $\sqrt{-3} \times \sqrt{-6}$

Options

- A. $-3\sqrt{2}$
- B. $3\sqrt{2}$
- C. $3\sqrt{2} i$
- D. $-3\sqrt{2} i$

Correct answer: $-3\sqrt{2}$

Explanation and steps

$\sqrt{-3} = i\sqrt{3}$ and $\sqrt{-6} = i\sqrt{6}$. Their product is $i^2\sqrt{18} = -3\sqrt{2}$.

Question 3 - Q3(i)

Simplify $3 + \sqrt{-64}$

and express it in the form $a + ib$

Options

- A. $3 + 8i$
- B. $3 - 8i$
- C. $11i$
- D. $3 + 64i$

Correct answer: $3 + 8i$

Explanation and steps

Use $i^2 = -1$, rationalise denominators with conjugates where needed, and collect real and imaginary parts. The result is $3 + 8i$.

Question 4 - Q3(ii)

Simplify $(2i^3)^2$
and express it in the form $a + ib$

Options

- A. $-4 + 0i$
- B. $4 + 0i$
- C. $-2i$
- D. $4i$

Correct answer: $-4 + 0i$

Explanation and steps

Use $i^2 = -1$, rationalise denominators with conjugates where needed, and collect real and imaginary parts. The result is $-4 + 0i$.

Question 5 - Q3(iii)

Simplify $(2 + 3i)(1 - 4i)$
and express it in the form $a + ib$

Options

- A. $14 - 5i$
- B. $14 + 5i$
- C. $-10 - 5i$
- D. $2 - 12i$

Correct answer: $14 - 5i$

Explanation and steps

Use $i^2 = -1$, rationalise denominators with conjugates where needed, and collect real and imaginary parts. The result is $14 - 5i$.

Question 6 - Q3(iv)

Simplify $(5/2)i(-4 - 3i)$
and express it in the form $a + ib$

Options

- A. $15/2 - 10i$
- B. $-15/2 - 10i$
- C. $15/2 + 10i$
- D. $-10 + (15/2)i$

Correct answer: $15/2 - 10i$

Explanation and steps

Use $i^2 = -1$, rationalise denominators with conjugates where needed, and collect real and imaginary parts. The result is $15/2 - 10i$.

Question 7 - Q3(v)

Simplify $(1 + 3i)^2(3 + i)$
and express it in the form $a + ib$

Options

- A. $-30 + 10i$
- B. $-30 - 10i$
- C. $30 + 10i$
- D. $-10 + 30i$

Correct answer: $-30 + 10i$

Explanation and steps

Use $i^2 = -1$, rationalise denominators with conjugates where needed, and collect real and imaginary parts. The result is $-30 + 10i$.

Question 8 - Q3(vi)

Simplify $(4 + 3i)/(1 - i)$
and express it in the form $a + ib$

Options

- A. $1/2 + (7/2)i$
- B. $1/2 - (7/2)i$
- C. $7/2 + (1/2)i$
- D. $-1/2 + (7/2)i$

Correct answer: $1/2 + (7/2)i$

Explanation and steps

Use $i^2 = -1$, rationalise denominators with conjugates where needed, and collect real and imaginary parts. The result is $1/2 + (7/2)i$.

Question 9 - Q3(vii)

Simplify $(1 + (2)/(i))(3 + (4)/(i))(5 + i)^{-1}$
and express it in the form $a + ib$

Options

- A. $-35/26 - (45/26)i$
- B. $-35/26 + (45/26)i$
- C. $35/26 - (45/26)i$
- D. $-45/26 - (35/26)i$

Correct answer: $-35/26 - (45/26)i$

Explanation and steps

Use $i^2 = -1$, rationalise denominators with conjugates where needed, and collect real and imaginary parts. The result is $-35/26 - (45/26)i$.

Question 10 - Q3(viii)

Simplify $(\sqrt{5} + \sqrt{3}i)/(\sqrt{5} - \sqrt{3}i)$
and express it in the form $a + ib$

Options

- A. $1/4 + (\sqrt{15}/4)i$
- B. $1/4 - (\sqrt{15}/4)i$
- C. $4 + \sqrt{15}i$
- D. $\sqrt{15}/4 + (1/4)i$

Correct answer: $1/4 + (\sqrt{15}/4)i$

Explanation and steps

Use $i^2 = -1$, rationalise denominators with conjugates where needed, and collect real and imaginary parts. The result is $1/4 + (\sqrt{15}/4)i$.

Question 11 - Q3(ix)

Simplify $(3i^5 + 2i^7 + i^9)/(i^6 + 2i^8 + 3i^{18})$
and express it in the form $a + ib$

Options

- A. $-i$
- B. i
- C. 1
- D. -1

Correct answer: $-i$

Explanation and steps

Use $i^2 = -1$, rationalise denominators with conjugates where needed, and collect real and imaginary parts. The result is $-i$.

Question 12 - Q3(x)

Simplify $(5 + 7i)/(4 + 3i) + (5 + 7i)/(4 - 3i)$
and express it in the form $a + ib$

Options

- A. $8/5 + (56/25)i$
- B. $8/5 - (56/25)i$
- C. $82/25 + 0i$
- D. $56/25 + (8/5)i$

Correct answer: $8/5 + (56/25)i$

Explanation and steps

Use $i^2 = -1$, rationalise denominators with conjugates where needed, and collect real and imaginary parts. The result is $8/5 + (56/25)i$.

Question 13 - Q4(i)

Solve $(4 - 5i)x + (2 + 3i)y = 10 - 7i$
for $x, y \in \mathbb{R}$

Options

- A. $x = 2, y = 1$
- B. $x = 1, y = 2$
- C. $x = 2, y = -1$
- D. $x = -2, y = 1$

Correct answer: $x = 2, y = 1$

Explanation and steps

Expand the complex equation and equate its real and imaginary parts. Solve the resulting two real equations to obtain $x = 2, y = 1$.

Question 14 - Q4(ii)

Solve $(1 - 3i)x + (2 + 5i)y = 7 + i$
for $x, y \in \mathbb{R}$

Options

- A. $x = 3, y = 2$
- B. $x = 2, y = 3$
- C. $x = -3, y = 2$
- D. $x = 3, y = -2$

Correct answer: $x = 3, y = 2$

Explanation and steps

Expand the complex equation and equate its real and imaginary parts. Solve the resulting two real equations to obtain $x = 3, y = 2$.

Question 15 - Q4(iii)

Solve $(x + iy)/(2 + 3i) = 7 - i$
for $x, y \in \mathbb{R}$

Options

- A. $x = 17, y = 19$
- B. $x = 19, y = 17$
- C. $x = 11, y = 23$
- D. $x = 17, y = -19$

Correct answer: $x = 17, y = 19$

Explanation and steps

Expand the complex equation and equate its real and imaginary parts. Solve the resulting two real equations to obtain $x = 17, y = 19$.

Question 16 - Q4(iv)

Solve $(x + iy)(5 + 6i) = 2 + 3i$
for $x, y \in \mathbb{R}$

Options

- A. $x = 28/61, y = 3/61$
- B. $x = 3/61, y = 28/61$
- C. $x = 28/61, y = -3/61$
- D. $x = 2/5, y = 1/2$

Correct answer: $x = 28/61, y = 3/61$

Explanation and steps

Expand the complex equation and equate its real and imaginary parts. Solve the resulting two real equations to obtain $x = 28/61, y = 3/61$.

Question 17 - Q4(v)

Solve $2x + i^9y(2 + i) = xi^7 + 10i^{16}$
for $x, y \in \mathbb{R}$

Options

- A. $x = 4, y = -2$
- B. $x = -4, y = 2$
- C. $x = 2, y = -4$
- D. $x = 4, y = 2$

Correct answer: $x = 4, y = -2$

Explanation and steps

Expand the complex equation and equate its real and imaginary parts. Solve the resulting two real equations to obtain $x = 4, y = -2$.

Question 18 - Q5(i)

Find $x^3 + 2x^2 - 3x + 21$
if $x = 1 + 2i$

Correct answer: 1

Explanation and steps

Substitute the given complex value of x , expand powers using $i^2 = -1$, and combine terms. The value is 1.

Question 19 - Q5(ii)

Find $x^3 - 5x^2 + 4x + 8$
if $x = (10)/(3 - i)$

Correct answer: -2

Explanation and steps

Substitute the given complex value of x , expand powers using $i^2 = -1$, and combine terms. The value is -2.

Question 20 - Q5(iii)

Find $x^3 - 3x^2 + 19x - 20$
if $x = 1 - 4i$

Correct answer: -3

Explanation and steps

Substitute the given complex value of x , expand powers using $i^2 = -1$, and combine terms. The value is -3.

Question 21 - Q6(i)

Find the square roots of $-16 + 30i$

Options

- A. $\pm(3 + 5i)$
- B. $\pm(3 - 5i)$
- C. $\pm(5 + 3i)$
- D. $\pm(4 + \sqrt{(30)} i)$

Correct answer: $\pm(3 + 5i)$

Explanation and steps

Set $a + ib$ as a square root, equate $a^2 - b^2$ and $2ab$ with the real and imaginary parts, then include both signs. The roots are $\pm(3 + 5i)$.

Question 22 - Q6(ii)

Find the square roots of $15 - 8i$

Options

- A. $\pm(4 - i)$
- B. $\pm(4 + i)$
- C. $\pm(1 - 4i)$
- D. $\pm(3 - 2i)$

Correct answer: $\pm(4 - i)$

Explanation and steps

Set $a + ib$ as a square root, equate $a^2 - b^2$ and $2ab$ with the real and imaginary parts, then include both signs. The roots are $\pm(4 - i)$.

Question 23 - Q6(iii)

Find the square roots of $2 + 2\sqrt{3}i$

Options

- A. $\pm(\sqrt{3} + i)$
- B. $\pm(\sqrt{3} - i)$
- C. $\pm(1 + \sqrt{3}i)$
- D. $\pm(2 + \sqrt{3}i)$

Correct answer: $\pm(\sqrt{3} + i)$

Explanation and steps

Set $a + ib$ as a square root, equate $a^2 - b^2$ and $2ab$ with the real and imaginary parts, then include both signs. The roots are $\pm(\sqrt{3} + i)$.

Question 24 - Q6(iv)

Find the square roots of $18i$

Options

- A. $\pm 3(1 + i)$
- B. $\pm 3(1 - i)$
- C. $\pm(9 + i)$
- D. $\pm 3i$

Correct answer: $\pm 3(1 + i)$

Explanation and steps

Set $a + ib$ as a square root, equate $a^2 - b^2$ and $2ab$ with the real and imaginary parts, then include both signs. The roots are $\pm 3(1 + i)$.

Question 25 - Q6(v)

Find the square roots of $3 - 4i$

Options

- A. $\pm(2 - i)$
- B. $\pm(2 + i)$
- C. $\pm(1 - 2i)$
- D. $\pm(3 - 4i)$

Correct answer: $\pm(2 - i)$

Explanation and steps

Set $a + ib$ as a square root, equate $a^2 - b^2$ and $2ab$ with the real and imaginary parts, then include both signs. The roots are $\pm(2 - i)$.

Question 26 - Q6(vi)

Find the square roots of $6 + 8i$

Options

A. $\pm\sqrt{2}(2 + i)$

B. $\pm\sqrt{2}(2 - i)$

C. $\pm(2 + 2i)$

D. $\pm(3 + i)$

Correct answer: $\pm\sqrt{2}(2 + i)$

Explanation and steps

Set $a + ib$ as a square root, equate $a^2 - b^2$ and $2ab$ with the real and imaginary parts, then include both signs. The roots are $\pm\sqrt{2}(2 + i)$.